

10 (**short**)

An isotropic elastic material with Young's modulus E and Poisson's ratio ν is stressed in its principal axes with plane strain condition (i.e., $\epsilon_3 = 0$).

(a) Using 3D Hooke's law, find an expression for σ_3 in terms of σ_1 and σ_2 , and express the matrix C such that

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} = C \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix}$$

in terms of E and ν .

[5]

(b) Find the relationship between ϵ_1 and ϵ_2 when $\nu = 0.5$. What is the physical interpretation of this equation?

[5]